भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab

Practice Assignment 4

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Consider the Iris species dataset regarding the classification of Iris flower species based different given predictors. Write a program to do the followings:
   (a) Randomly split the dataset into training and testing splits using k-fold cross validation, for k=5.
   Hint: Consider 10% of the training set as validation set.
   (b) Design the hypothesis ($\hat{y}$) to classify the Iris species using Multi-class Logistic regression to estimation of the model parameters (θ) or (ω) for the different training-testing splits obtained from 1(b). Apply the followings for this:
   - Batch gradient descent method for calculating error-gradient.
   - One-vs-all strategy for multi-class scenario.

   without using in-built Python packages /libraries for both.
   - Consider model parameters are to be initialized (θ) or (ω) to zero.
   - Apply hyper-parameter tuning for selecting the best model for the following hyper-parameters, considering the given values:
     1. learning rate, α: $\{0.0001, 0.1\}$
     2. ρ: $\{.0.001, 0.01\}$ where, ρ signifies absolute error-difference in two consecutive epochs.
     3. epoch , (T): $\{50,100\}$

   (c) Calculate and visualize the confusion matrix, considering the designed hypothesis (($\hat{y}$) and the actual targets (y), for all the training: testing splits.
   (d) Calculate and report the overall accuracy, class-wise accuracy, precision and recall values, with respect to average of the k-folds and the best fold, in separate sheets, and also write the result analysis.

2. Download Consider the Wine quality dataset regarding 13 constituents present in wine, without the information of the wine types (target). Write a program to do the followings:
   (a) Read the dataset.
   (b) Cluster the dataset into different categories of wine using K-means clustering method (**without using in-built Python library/ packages**). Consider the following for this:
   - Number of initial centroids, K to be 3, 5, 10, and 15.
     Hint: Apply K-means clustering, separately for all the given K-values.
   - Stopping criteria: No changes in the cluster assignment.

   (c) Report and visualize the obtained final clusters with respect to different K, and write the result analysis.